

RAG Chatbot: Complete README.md & Supplemental Files

Ready-to-use documentation file for your GitHub repository.

1. README.md File Contents

```
# RAG Document Chatbot 🤖
```

A lightweight, high-performance Retrieval-Augmented Generation (RAG) document chatbot built with Streamlit, Hugging Face Sentence Transformers, FAISS, and the official Groq API.

Features

- 📄 **Multi-Format Support**: Upload PDF and TXT files.
- ⚡ **Ultra-Fast LLM Inference**: Powered by Groq (`llama-3.3-70b-versatile`).
- 🧠 **Local Vector Search**: Runs `faiss-cpu` locally without external vector database fees.
- 🎯 **Prompt Grounding**: Answers questions strictly using retrieved context.
- 📌 **Source Citations**: Displays exact source file, page number, and chunk text.
- 🛡️ **Privacy & Security**: Secure handling of API keys via Streamlit Secrets.

Setup & Running Locally

1. Clone the Repository:

```
```bash
git clone https://github.com/YOUR_USERNAME/rag-streamlit-chatbot.git
cd rag-streamlit-chatbot
```
```

2. Create a Virtual Environment:

```
```bash
python -m venv venv
source venv/bin/activate # On Windows use: venv\Scripts\activate
```
```

3. Install Dependencies:

```
```bash
pip install -r requirements.txt
```
```

4. Configure API Key:

```
Create a file `.streamlit/secrets.toml` with:
```toml
GROQ_API_KEY = "gsk_your_actual_groq_api_key_here"
```
```

5. Run the Streamlit App:

```
```bash
streamlit run app.py
```
```

Limitations & Future Enhancements

- Scanned PDF images require an OCR engine (such as Tesseract) to extract text.
- FAISS vectors run in session RAM and reset upon browser tab refresh.

2. Complete File Summary List

In total, your project consists of 5 files:

- `app.py` — Main application logic (Included in Source Code PDF) [cite: 1]
- `requirements.txt` — Python dependencies (Included in Source Code PDF) [cite: 1]
- `.gitignore` — Git ignore list (Included in Source Code PDF) [cite: 1]
- `README.md` — GitHub project documentation (Included in this PDF)
- `.streamlit/config.toml` — Streamlit UI configuration (Included in Source Code PDF) [cite: 1]